TIRTH BHAVINKUMAR POPAT

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EDUCATION

Vellore Institute of Technology (VIT) — **Chennai, India**
B.Tech Electronics Engineering – VLSI Design and Technology, ***CGPA: 8.74*** — ***July 2024 – Present***

Puna International School — **Ahmedabad, India**
Class XII CBSE – ***86%****, JEE Advanced Rank:* ***18791*** — ***April 2024***

Podar International School — **Ahmedabad, India**
Class X CBSE – ***88%*** — ***July 2022***

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, MATLAB, Verilog HDL, R
Tools: Xilinx Vivado, Quartus Prime, LTspice, MATLAB, Keil, Arduino IDE, Processing IDE, COMSOL Multiphysics
Soft Skills: Communication, Leadership, Team Collaboration, Problem Solving

RESEARCH PUBLICATIONS & MANUSCRIPTS

A Low-Cost Digital Stethoscope with Machine Learning-Based Cardiac Sound Classification
- Presented at IEEE INDISCON 2026. Scheduled for publication in the IEEE Xplore Digital Library.

Isospectrality of Rationally Extended Quantum Harmonic Oscillators under Parity Time (PT) Symmetric Perturbations
- Manuscript under review.

EXPERIENCE

Summer Internship – VLSI Design — **May 2025 – Aug 2025**
Centre for Nano Electronics & VLSI Design (CNVD), VIT Chennai — ***Chennai, India***

- Designed and verified **5+ MAC and multiplier architectures** in Verilog HDL for DSP and AI/ML accelerators.
- Implemented an **8-bit signed Booth multiplier** with a 16-bit output using Xilinx Vivado.
- Performed **RTL simulation, synthesis, FPGA implementation, timing and power analysis** to evaluate area and performance trade-offs.

PROJECTS

Digital Stethoscope System with ML-based Heart Sound Classification

Python, Signal Processing, 1D-CNN, MFCC | **Oct 2025 – Sept 2026**

- Developed a **low-cost digital stethoscope** integrating PCG signal acquisition, noise filtering, segmentation, and **1D-CNN classification** of Normal, Murmur, and Artifact heart sounds.
- Processed **520 heart sound recordings** using **120 MFCC-based features per frame** across 200-frame sequences, achieving **89.42% overall accuracy** and **90.04% abnormal-sound accuracy**.
- Implemented **50 Hz notch filtering**, sixth-order Butterworth low-pass filtering, and feature extraction using **MFCCs, delta, and delta-delta coefficients** for robust PCG signal analysis.

CPW Resonator Automation

Python, Microwave Engineering, CadQuery, COMSOL Multiphysics | **Jul 2026 – Oct 2026**

- Written a **Python-based automation framework** for designing **quarter-wave CPW resonators**, converting user-defined RF parameters into resonator designs.
- Automated calculations of **4 key RF parameters**: characteristic impedance, effective dielectric constant, guided wavelength, and resonator length.
- Generated parametric **3D CAD models** of the substrate, ground plane, feed line, and resonator using **CadQuery**, supporting **4 export formats**: STEP, STL, BREP, and JSON, with a workflow for **COMSOL Multiphysics** simulation.

Drift-Sense: ML-based Navigation-Error Recovery for Wafer Inspection Tools

Python, PyTorch, OpenCV, Siamese ResNet-18, ZNCC | **Aug 2026 – Present**

- Formed a hybrid **Siamese ResNet-18 + Rotated ZNCC** pipeline for precise localization of semiconductor wafer features in SEM inspection images under scale, rotation, noise, and imaging variations.
- Generated synthetic **FinFET SEM datasets** with realistic noise, blur, drift, rotation, and imaging artefacts to train and evaluate the localization pipeline without proprietary fab data.
- Achieved **100% localization accuracy under no noise, 96.7% under low noise, and 90.0% under high noise** at the 5-pixel tolerance threshold across 90 curated test samples.
- Achieved mean localization errors of **0.397 px, 2.305 px, and 4.518 px** under no-, low-, and high-noise conditions, respectively.

CERTIFICATIONS

C-DAC Bengaluru

Quantum Computing Training – Qniverse Platform — **Feb 2026** — ***Bengaluru, India***

- Trained in **quantum computing fundamentals**, including qubits, quantum gates, and circuits, using the Qniverse simulation platform, with a visit to C-DAC's Quantum Photonics Laboratory and Supercomputing facilities.

VIT Chennai – School of Electronics Engineering (SENSE)

Summer Workshop – Digital VLSI System Design — **May 2025 – Jun 2025** — ***Chennai, India***

- Implemented and verified **combinational and sequential circuits** using **Verilog HDL**, including multiplexers, decoders, adders, Gray-to-Binary converters, flip-flops, FSM-based sequence detectors, and counters, with simulation testbenches.

COURSEWORK

- Complex Variables and Linear Algebra **S**
- Probability and Statistics **S**
- Signal Processing **A**
- Electromagnetic Field Theory **A**
- Electronic Materials **A**
- Digital Systems Design **A**
- Electronic Circuits **A**
- Quantum Technology for Electronics Engineers **A**